

EG060V-EA

Log Capture Guide

LTE-A Module Series

Version: 1.0

Date: 2021-12-21

Status: Released



At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

<http://www.quectel.com/support/sales.htm>.

For technical support, or to report documentation errors, please visit:

<http://www.quectel.com/support/technical.htm>.

Or email us at: support@quectel.com.

Legal Notices

We offer information as a service to you. The provided information is based on your requirements and we make every effort to ensure its quality. You agree that you are responsible for using independent analysis and evaluation in designing intended products, and we provide reference designs for illustrative purposes only. Before using any hardware, software or service guided by this document, please read this notice carefully. Even though we employ commercially reasonable efforts to provide the best possible experience, you hereby acknowledge and agree that this document and related services hereunder are provided to you on an “as available” basis. We may revise or restate this document from time to time at our sole discretion without any prior notice to you.

Use and Disclosure Restrictions

License Agreements

Documents and information provided by us shall be kept confidential, unless specific permission is granted. They shall not be accessed or used for any purpose except as expressly provided herein.

Copyright

Our and third-party products hereunder may contain copyrighted material. Such copyrighted material shall not be copied, reproduced, distributed, merged, published, translated, or modified without prior written consent. We and the third party have exclusive rights over copyrighted material. No license shall be granted or conveyed under any patents, copyrights, trademarks, or service mark rights. To avoid ambiguities, purchasing in any form cannot be deemed as granting a license other than the normal non-exclusive, royalty-free license to use the material. We reserve the right to take legal action for noncompliance with abovementioned requirements, unauthorized use, or other illegal or malicious use of the material.

Trademarks

Except as otherwise set forth herein, nothing in this document shall be construed as conferring any rights to use any trademark, trade name or name, abbreviation, or counterfeit product thereof owned by Quectel or any third party in advertising, publicity, or other aspects.

Third-Party Rights

This document may refer to hardware, software and/or documentation owned by one or more third parties ("third-party materials"). Use of such third-party materials shall be governed by all restrictions and obligations applicable thereto.

We make no warranty or representation, either express or implied, regarding the third-party materials, including but not limited to any implied or statutory, warranties of merchantability or fitness for a particular purpose, quiet enjoyment, system integration, information accuracy, and non-infringement of any third-party intellectual property rights with regard to the licensed technology or use thereof. Nothing herein constitutes a representation or warranty by us to either develop, enhance, modify, distribute, market, sell, offer for sale, or otherwise maintain production of any our products or any other hardware, software, device, tool, information, or product. We moreover disclaim any and all warranties arising from the course of dealing or usage of trade.

Privacy Policy

To implement module functionality, certain device data are uploaded to Quectel's or third-party's servers, including carriers, chipset suppliers or customer-designated servers. Quectel, strictly abiding by the relevant laws and regulations, shall retain, use, disclose or otherwise process relevant data for the purpose of performing the service only or as permitted by applicable laws. Before data interaction with third parties, please be informed of their privacy and data security policy.

Disclaimer

- a) We acknowledge no liability for any injury or damage arising from the reliance upon the information.
- b) We shall bear no liability resulting from any inaccuracies or omissions, or from the use of the information contained herein.
- c) While we have made every effort to ensure that the functions and features under development are free from errors, it is possible that they could contain errors, inaccuracies, and omissions. Unless otherwise provided by valid agreement, we make no warranties of any kind, either implied or express, and exclude all liability for any loss or damage suffered in connection with the use of features and functions under development, to the maximum extent permitted by law, regardless of whether such loss or damage may have been foreseeable.
- d) We are not responsible for the accessibility, safety, accuracy, availability, legality, or completeness of information, advertising, commercial offers, products, services, and materials on third-party websites and third-party resources.

Copyright © Quectel Wireless Solutions Co., Ltd. 2021. All rights reserved.

About the Document

Revision History

Version	Date	Author	Description
-	2021-11-10	Paddy LI	Creation of the document
1.0	2021-12-21	Paddy LI	First official release

Contents

About the Document.....	3
Contents.....	4
Table Index.....	5
Figure Index.....	6
1 Introduction	7
2 Capture Log and Dump File in Windows.....	8
2.1. Capture Log	8
2.1.1. Capture AP Log Through Serial Port Tool.....	8
2.1.2. Capture CP Log Through CATStudio.....	9
2.2. Capture Dump File	15
3 Capture Log and Dump File in Linux.....	16
3.1. Capture Log	16
3.1.1. Capture AP Log Through Serial Port Tool.....	16
3.1.2. Capture CP Log Through QLog (Log Saved in Linux).....	16
3.1.3. Capture CP Log Through QLog (Log Saved in Windows).....	18
3.2. Capture Dump File	19
4 Dump Mode Related AT Commands.....	22
4.1. AT Command Introduction	22
4.1.1. Definitions.....	22
4.1.2. AT Command Syntax	22
4.2. AT Command Description	23
4.2.1. AT+QCFG="aprstlevel" Set AP Dump Level	23
4.2.2. AT+QCFG="modemrstlevel" Set CP Dump Level.....	23
5 Network State Checking Related AT Commands.....	25
6 Appendix References	26

Table Index

Table 1: Types of AT Commands	22
Table 2: Related Documents	26
Table 3: Terms and Abbreviations.....	26

Figure Index

Figure 1: Boot Log of the Module	8
Figure 2: CATStudio Mode Selection Interface	9
Figure 3: CATStudio Main Interface.....	10
Figure 4: Communication Settings Interface	10
Figure 5: Logger Operation Interface	11
Figure 6: Database Selection Interface	12
Figure 7: Logger Interface Presented After Successful Database Pairing	12
Figure 8: CATStudio Main Interface Tool Bar	13
Figure 9: LogViewer Interface	13
Figure 10: Tool Bar Interface to Export Log	14
Figure 11: ExportLog Interface to Select Log Export Path	14
Figure 12: AP Dump File Exported When AP Dumps.....	15
Figure 13: CP Dump File Exported When CP Dumps.....	15
Figure 14: Loaded Module Ports	16
Figure 15: Capture CP Log on Linux Device	17
Figure 16: View CP Log on Linux Device	17
Figure 17: Socket Port Selection Interface	18
Figure 18: Connection Method of Capturing Dump File in Linux System	19
Figure 19: NIC Setting of Windows PC	19
Figure 20: Printed Information After Successful Capture of Log by rdp_transfer	21

1 Introduction

This document mainly introduces how to capture AP log, CP log and dump file on Quectel LTE-A EG060V-EA module through QLog, rdp_transfer, CATStudio and TFTP32.

When the module is in an abnormal condition, you can check the current network state of the module through AT commands (see **Chapter 5** for details) to determine the cause of the problem based on the query results. If it is determined that the module is misconfigured, you can reconfigure the module with the corresponding AT command to resolve the problem. If the cause cannot be determined or the problem cannot be resolved, you can capture corresponding logs and send the logs to Quectel for further analysis and processing.

NOTE

1. QLog is a log capture tool developed by Quectel, you can contact Quectel Technical Supports to obtain the installation package if necessary.
2. You can install rdp_transfer and CATStudio tool with the assistance of Quectel Technical Supports.
3. You can download TFTP32 from official website: <https://bitbucket.org/phjounin/tftp32/downloads/>.

2 Capture Log and Dump File in Windows

2.1. Capture Log

2.1.1. Capture AP Log Through Serial Port Tool

Step 1: Connect the module to the host through the debug UART port using a serial port line. Open any serial port tool on the host, select the COM port loaded in the device manager window and set the baud rate to 115200 bps. Then disable flow control and power on the module. The boot log will be displayed in the serial port tool.

```
PowerDown log register2:0xe1
-----
| name(power down2) | status |
|-----|-----|
| HYB_DONE          | 1      |
| UV_VSYS2          | 0      |
| HW_RESET          | 0      |
| PG00D_PDOWN       | 0      |
| LONKEY_RTC         | 0      |
| RSVD              | 1      |
|-----|-----|
smpl_reboot_flag: 0
CTRL+C: enter instruction mode
PINTEST OR FASTBOOT

NAND: detect chip id 61c8 on cs 0, ESMT 1Gibx8
Max capacity of BBM is 54 blocks!!
[abbt] relocation table at page:63
The reloc table is empty now
[bbt] relocation table at page:63
The reloc table is empty now
*** Warning - bad CRC, using default environment

In: serial
Out: serial
Err: serial
The SVC profile is 8
The CA7 is running at OP4,DVL:3
The AXI is running at OP3,DVL:3
The DDR is running at OP4,DVL:3
Got ASR Flag Offset: 0x5f00000.
@@ Retrive image offset/size from TIM.
@@ Default AB flags: active_slot: a, reboot_cnt: 0, synced: 1
@@ DTIM@0x100000, ASR FLAG@0x5f00000
@@ Retrive image offset/size from DTIM.
@@ Uboot@0x1f80000, Kernel@0x2100000, Rootfs@0x2b00000, CP(Arbel@0x5c0000, Msa@0x1c0000, RF@0x180000)
@@Active System is A.
```

Figure 1: Boot Log of the Module

Step 2: Tap “Enter” on the keyboard and then an "OpenWrt login:" window will be popped up. Enter username “boot” and password “admin” in the "OpenWrt login:" window to log in to the Linux system of the module. AP log contains kernel log and API log of QLog interface. Input **echo 1 > /sys/module/printk/parameters/kernel_log_output** at the debug port through the serial port tool to print kernel log; input **logcat -b radio** at the debug port through the serial port tool to print the API log of QLog interface.

2.1.2. Capture CP Log Through CATStudio

In Windows system, the CP log of the module can be captured through the Diag port loaded by USB enumeration.

Step 1: Open CATStudio on the host, select “Generic Target Online” and then click “OK”.

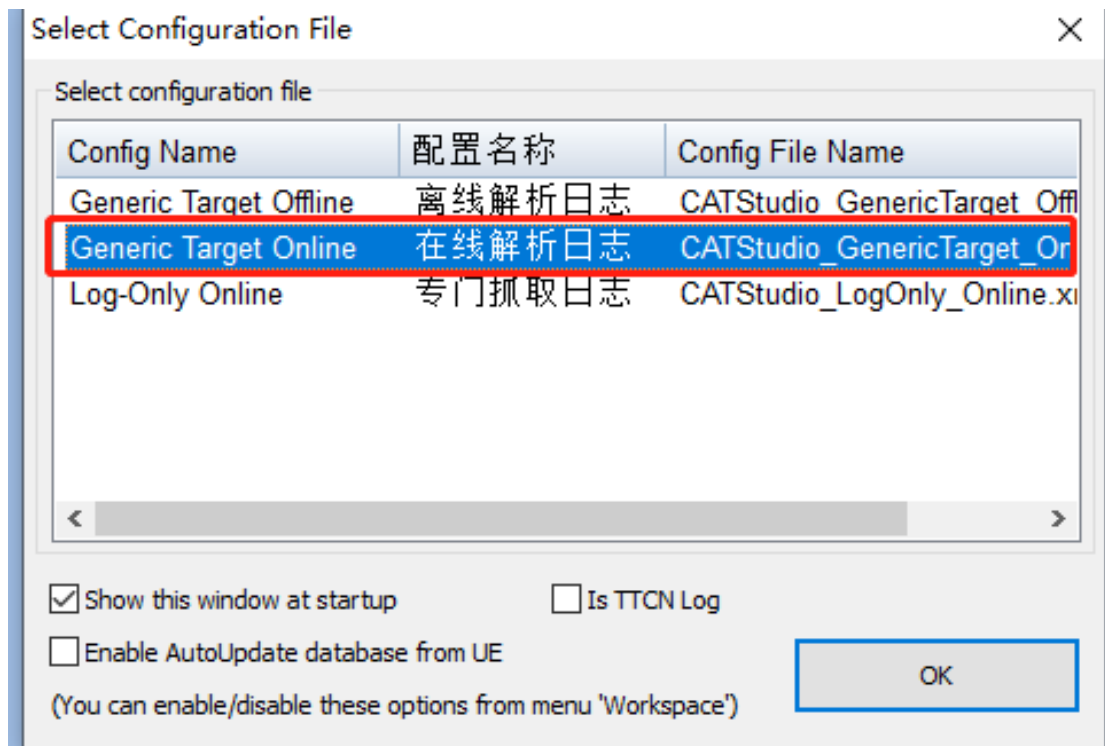


Figure 2: CATStudio Mode Selection Interface

Step 2: Select “Device Communication” on the lower right corner of the CATStudio main interface. On “Device Communication” interface, select “Settings” corresponding to “Device0”. Select “Quectel USB DIAG Port” on “Communication Settings” interface and set baud rate to 115200 bps, and then click “OK”.

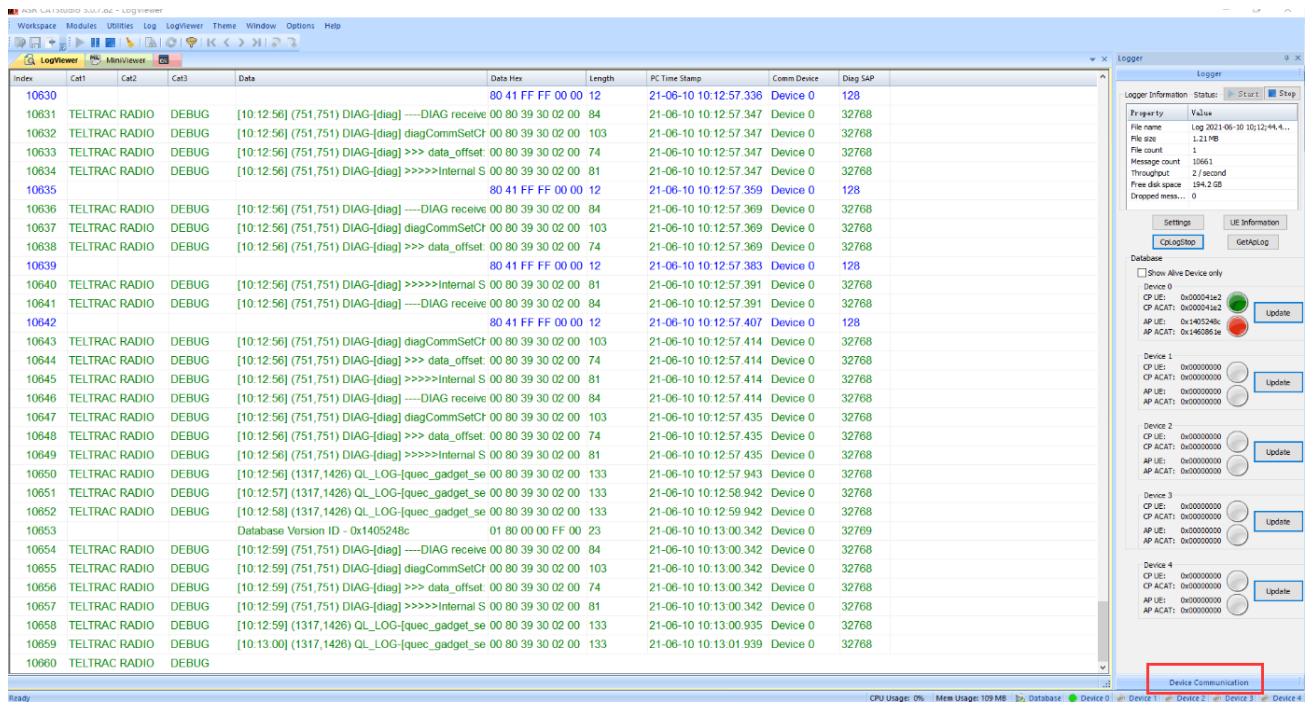


Figure 3: CATStudio Main Interface

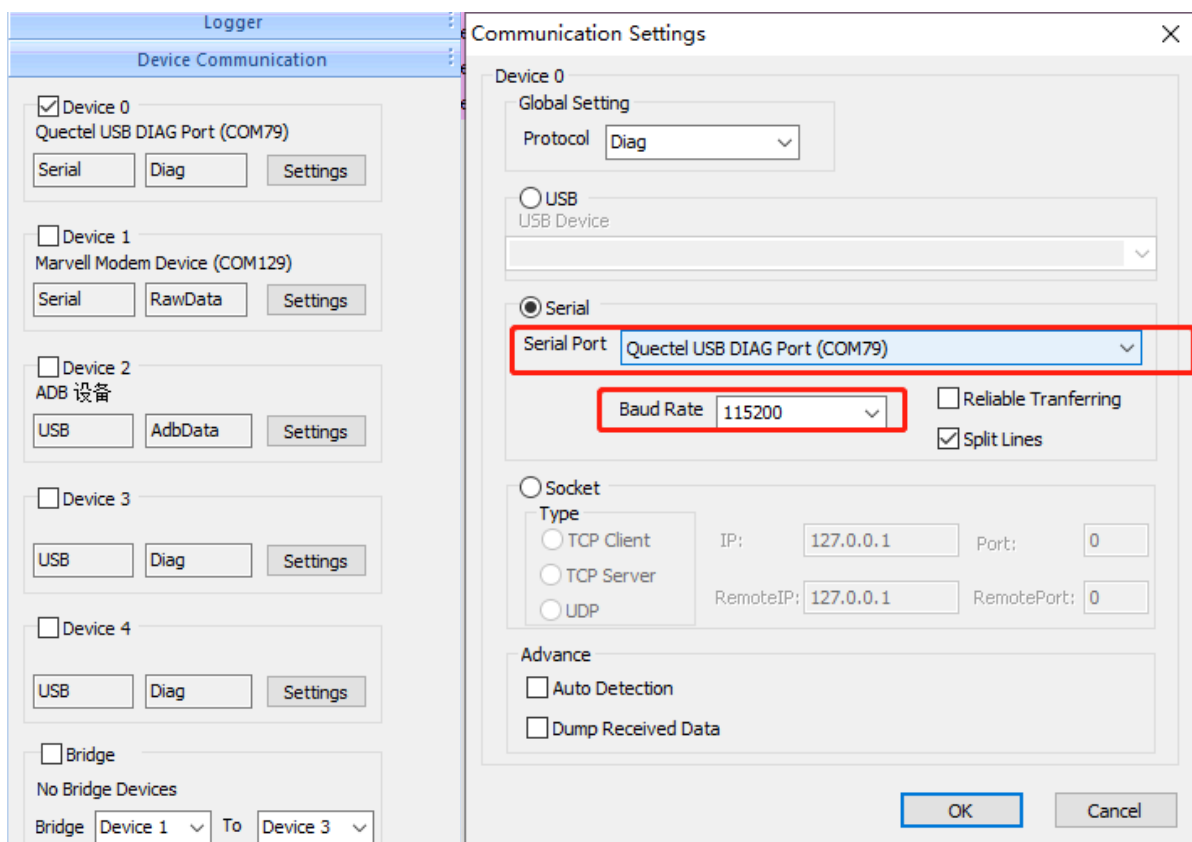


Figure 4: Communication Settings Interface

Step 3: Continue to select the “**Logger**” tab on main interface, click “**CpLog Start**” and then “**CpLogStart**” changes into “**CpLogStop**”. Then click “**Update**” in “**Database**” section to enter the “**Database**” selection interface.

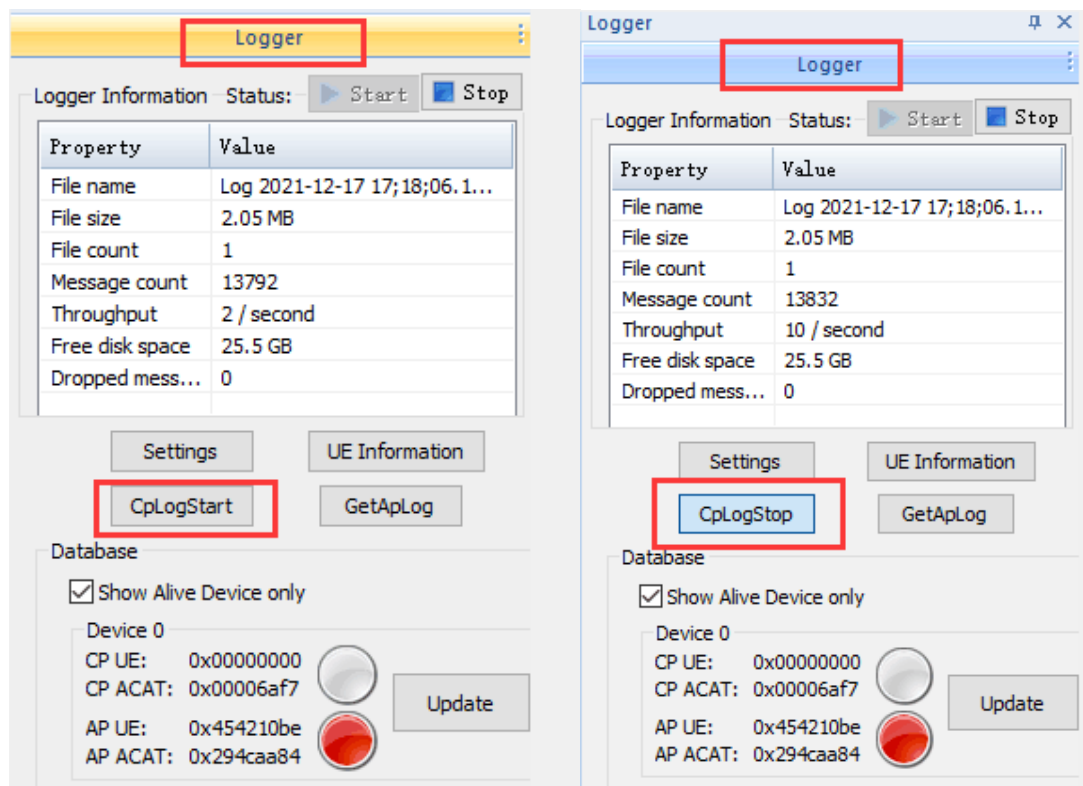


Figure 5: Logger Operation Interface

Step 4: On “**Database**” selection interface, select “**Text**” in “**Communication**” section, and then select *LWG_MDB.txt* in “**Text file**”; select “**Text**” in “**Application**” section, and then select *Boerne_DIAG.mdb.txt* in “**Text file**”, then click “**UpdateAll**”. When the updating process completes, close the “**Database**” selection interface and return to the “**Logger**” interface, then two indicators turn green.

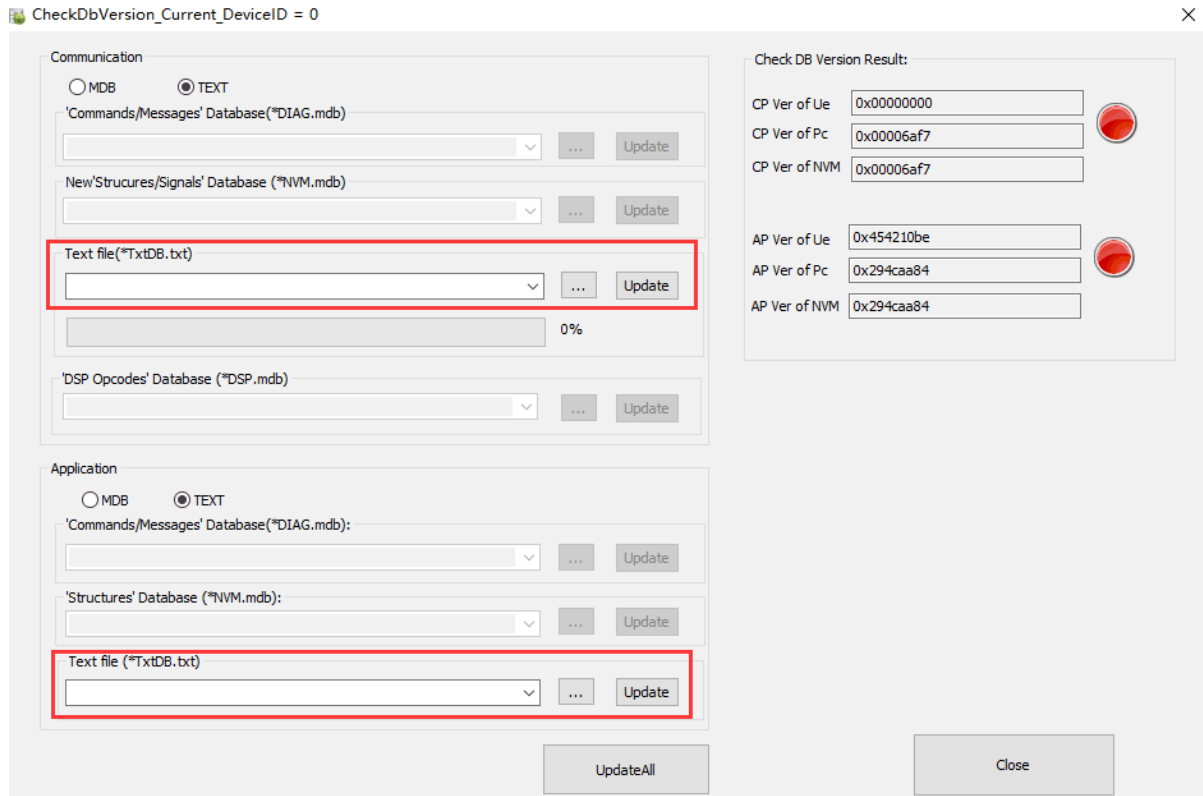


Figure 6: Database Selection Interface

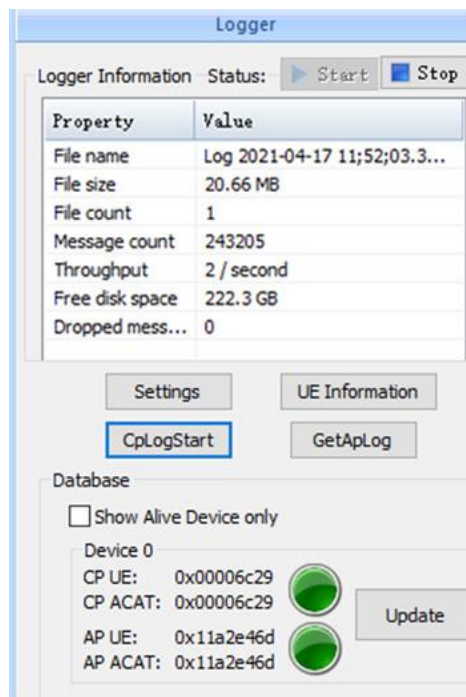


Figure 7: Logger Interface Presented After Successful Database Pairing

Step 5: Select “Modules” in the tool bar of the main interface, then click “LogViewer” to open the log capture interface.

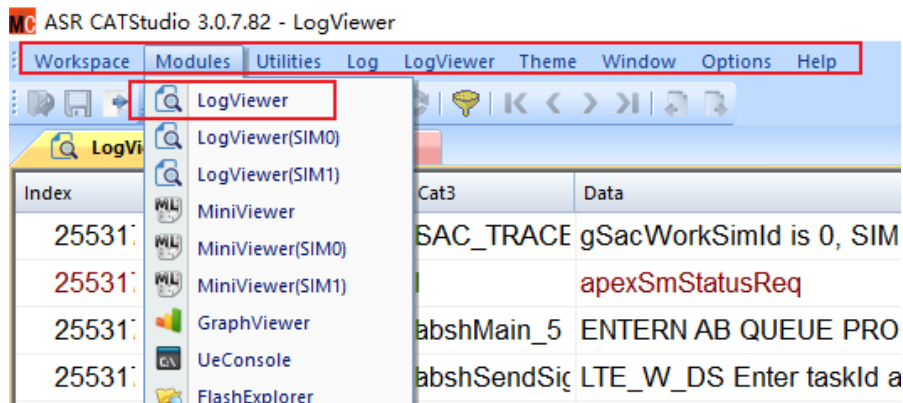


Figure 8: CATStudio Main Interface Tool Bar

LogViewer	MiniViewer						
Index	Cat1	Cat2	Cat3	Data	Data Hex	Length	PC Time
1870078	APLP	TM	pTmMain	pTmMain - event: (00913000) : active mode: TM_NO_SLEEP_MODE active state: 0001	00 00 E9 0D 0E 00 19	21-04	
1870079	APLP	MCL	pMODEMat	MCL: chain:3 mode:0 state:1 received event:0x00913000 - change state to:0	00 00 E9 0D 0E 00 24	21-04	
1870080	APLP	TM	pTmTracker	setActiveCellActionAck: {WbSetActiveCellActionAck}	00 00 E9 0D 0E 00 52	21-04	
1870081	APLP	TM	pTmStoreEc	pTmStoreEcNoAndRssiValue () - cellEcNoBitmap: 0001	00 00 E9 0D 0E 00 14	21-04	
1870082	APLP	TM	pTmStoreEc	pTmStoreEcNoAndRssiValue () - cellEcNoBitmap: 0001	00 00 E9 0D 0E 00 584	21-04	
1870083	APLP	TM	setSpecialM	set specialMaintenanceFlg:0	00 00 E9 0D 0E 00 14	21-04	
1870084	APLP	TM	pTmRestore	pTmRestoreMsgs	00 00 E9 0D 0E 00 12	21-04	
1870085	APLP	MS	getDrxTaskS	drxCtr: get drxTask DRX_RELEASE_PICH as DRX_TASK_COMPLETE	00 00 E9 0D 0E 00 14	21-04	
1870086	APLP	MAIN_LOOP	MainLoop	MainLoop - event: (b101e0ff) : active mode: MSR_IDLE_MODE active state: 0009	00 00 E9 0D 0E 00 27	21-04	
1870087	APLP	MSR	pMSRMain	pMSRMain - event: (b101e0ff) : active mode: MSR_IDLE_MODE active state: 0009	00 00 E9 0D 0E 00 19	21-04	
1870088	APLP	MCL	pMODEMat	MCL: chain:MSR_CHAIN mode:3 state:9 received event:0xb101e0ff - change state to:10	00 00 E9 0D 0E 00 23	21-04	
1870089	APLP	ApplExtTr	pMSrSetTr	Idle mode state set to: IFR_MEAS_IDLE_STARTING	00 00 E9 0D 0E 00 13	21-04	
1870090	APLP	ApplExtTr	pMSrInvoke	pMSrInvokeNextIfMeasIdle: function called at state: IFR_MEAS_IDLE_STARTING	00 00 E9 0D 0E 00 13	21-04	
1870091	APLP	MSR	pMSrDcxoC	pMSrDcxoC pMSrDcxoCheckAfcForIdleMeasurements callback 0x00000000 currentTime -7530 DcxoDb.LastDcxoRefTime 0 timePassed -7530	00 00 E9 0D 0E 00 22	21-04	
1870092	APLP	MSR	TIME_REM	time remained until end of measurements 1232 , piRefTimeValidity:1	00 00 E9 0D 0E 00 18	21-04	
1870093	APLP	MSR	TIME_REM	time remained until end of measurements 1232 , piRefTimeValidity:1	00 00 E9 0D 0E 00 18	21-04	
1870094	APLP	MSR	pMSrInvoke	ucfIdleModeState = 0x00000001 , interFreqBssFlag = 0 , interFreqMpsFlag = 0 , timeToMeasure = 1232	00 00 E9 0D 0E 00 24	21-04	
1870095	APLP	MSR	pMSrSetPer	pendingInterMeas set to value: 0	00 00 E9 0D 0E 00 14	21-04	
1870096	TELTRAC	RADIO	DEBUG	[12:55:28] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	
1870097	TELTRAC	RADIO	DEBUG	[12:55:29] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	
1870098	TELTRAC	RADIO	DEBUG	[12:55:30] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	
1870099				Database Version ID - 0x15b92ff3	01 80 00 00 FF 00 23	21-04	
1870100	TELTRAC	RADIO	DEBUG	[12:55:30] (742,742) DIAG-[diag] ----DIAG received from USB:14, len:16	00 80 FF FF 02 00 84	21-04	
1870101	TELTRAC	RADIO	DEBUG	[12:55:30] (742,742) DIAG-[diag] diagCommSetChunk total_expect:16 total_recv:16 current:16	00 80 FF FF 02 00 103	21-04	
1870102	TELTRAC	RADIO	DEBUG	[12:55:30] (742,742) DIAG-[diag] >>> data_offset:4, SAP:128	00 80 FF FF 02 00 74	21-04	
1870103	TELTRAC	RADIO	DEBUG	[12:55:30] (742,742) DIAG-[diag] >>>>Internal Service: Service 13	00 80 FF FF 02 00 81	21-04	
1870104	TELTRAC	RADIO	DEBUG	[12:55:31] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	
1870105	TELTRAC	RADIO	DEBUG	[12:55:32] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	
1870106	TELTRAC	RADIO	DEBUG	[12:55:33] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	
1870107	TELTRAC	RADIO	DEBUG	[12:55:34] (1109,1234) QL_LOG-[quec_gadget_ser] [quec_data_register:380] [quectel][net_status] exit QL_NET_STATUS_DATA	00 80 FF FF 02 00 133	21-04	

Figure 9: LogViewer Interface

Step 6: After reproducing the problem, click “Log” in the tool bar of main interface and select “Export Log-File...” in the drop-down box. In the “ExportLog” window popped up, set a custom path to save log files, and rename the log files to be exported, then click “Output(O)” to export the log files.

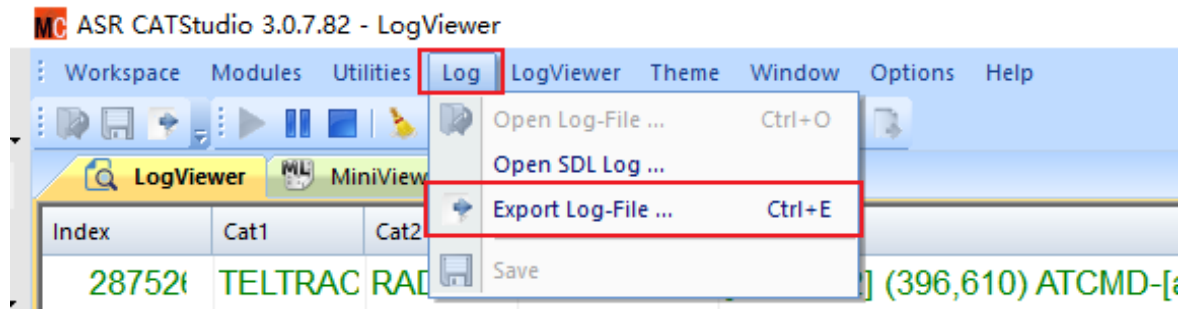


Figure 10: Tool Bar Interface to Export Log

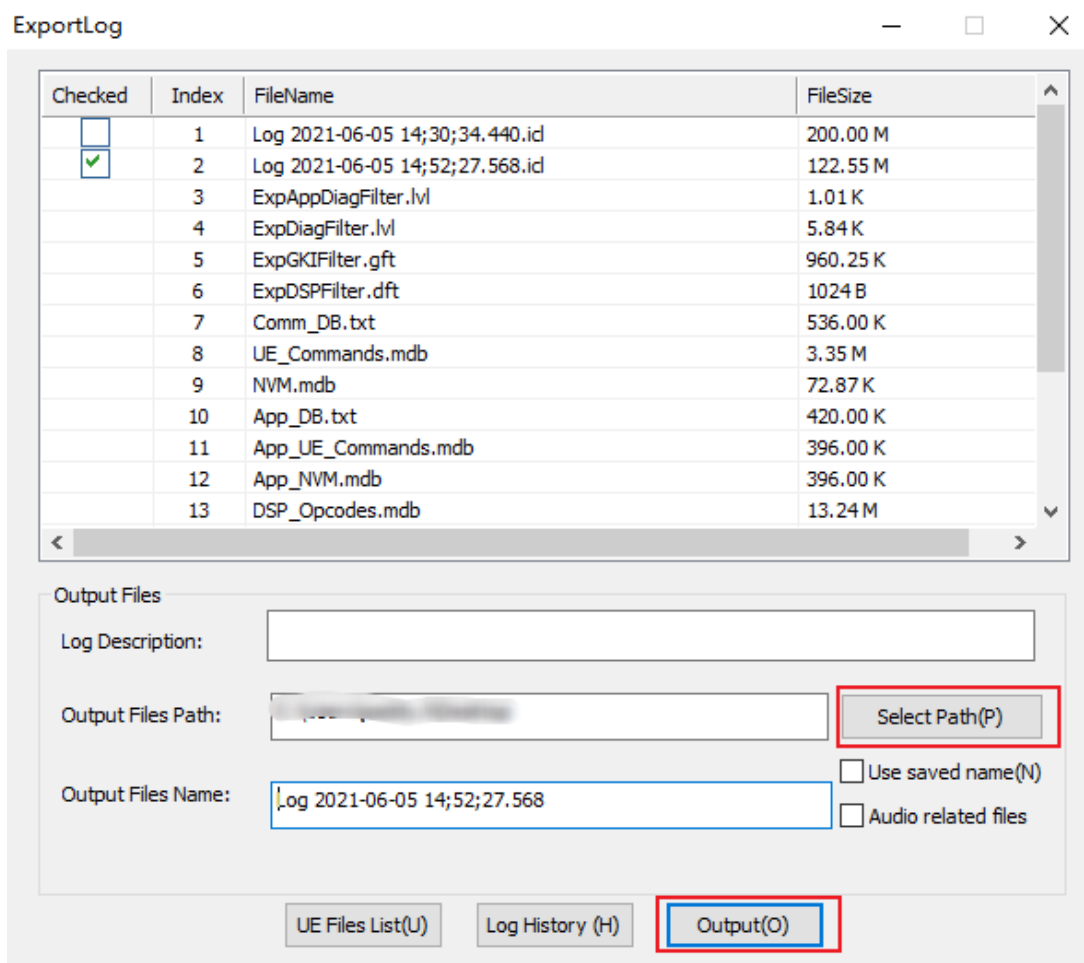


Figure 11: ExportLog Interface to Select Log Export Path

NOTE

LWG_MDB.txt and Boerne_DIAG.mdb.txt are database files, you can contact Quectel Technical Supports (Website: support@quectel.com) to obtain the files.

2.2. Capture Dump File

In Windows system, the dump file generated when the module dumps can be exported through TFTP32 in Windows. You can send the dump file to Quectel for further analysis.

Step 1: Configure the module into dump mode by executing **AT+QCFG="aprstlevel",0** and **AT+QCFG="modemrstlevel",0**. You have to reboot the module after **OK** is returned for both commands such that the commands can take effect. See **Chapter 4** for more details about the commands mentioned above.

Step 1: Open TFTP32 in Windows. The module will automatically reboot after it dumps and automatically executes **rdp_transfer app**. When the execution completes, TFTP32 will automatically export the dump file.

- If the AP dumps, the dump file exported by TFTP32 will contain a compressed package and a txt file. The txt file briefly describes the location of this dump to illustrate the problem more clearly.

 RAMDUMP_200925-1607.gz	2020/9/25 16:10
 RAMDUMP_200925-1607.txt	2020/9/25 16:08

Figure 12: AP Dump File Exported When AP Dumps

- If the CP dumps, the dump file exported by TFTP32 will contain two bin files and a compressed package.

 CPASSERT_DSPDDR_2020_12_28_19_21_35.bin	2020/12/28 19:21
 CPASSERT_COMDDR_2020_12_28_19_21_35.bin	2020/12/28 19:21
 CPASSERT_LOG_2020_12_28_19_21_35.tar.gz	2020/12/28 19:21

Figure 13: CP Dump File Exported When CP Dumps

3 Capture Log and Dump File in Linux

3.1. Capture Log

3.1.1. Capture AP Log Through Serial Port Tool

The steps for capturing AP log in Linux system are the same as those in Windows system. See **Chapter 2.1.1** for more details.

3.1.2. Capture CP Log Through QLog (Log Saved in Linux)

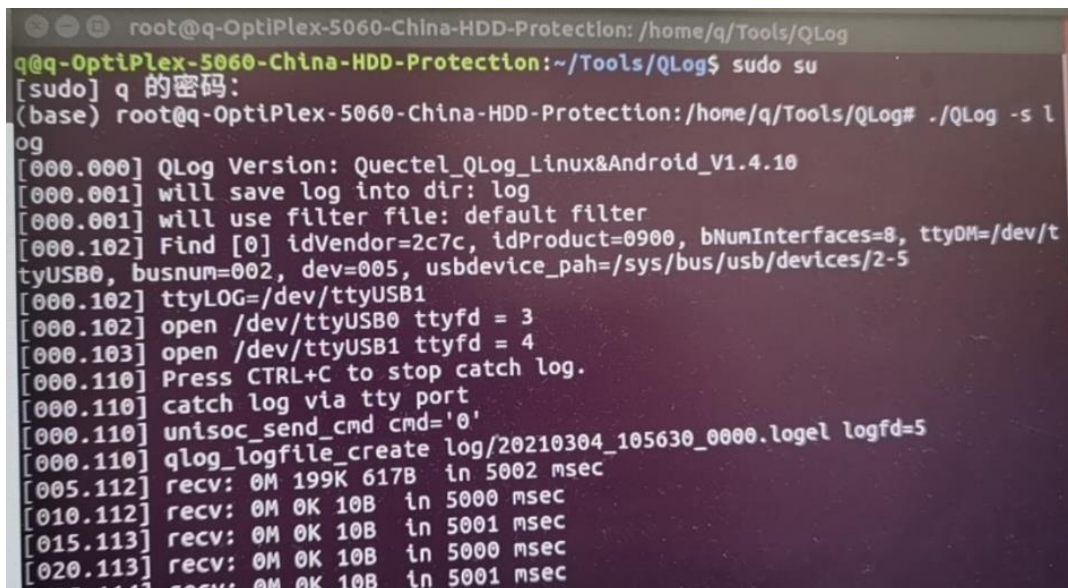
Step 1: Install Linux USB driver. You can contact Quectel Technical Supports to obtain the installation method.

Step 2: Open Linux terminal window under the root directory of QLog. Execute **make** to generate an executable file named *Qlog* under the QLog directory. Execute **dmesg -w** in Linux terminal window to list the module ports. If no port is displayed when it is determined that the module is connected, then the USB driver may be failed to be installed and needs to be reinstalled. If the module ports are loaded successfully, then QLog will start to capture log. The following information will be printed after the ports are loaded successfully.

```
[620635.180720] usb 1-1: new high-speed USB device number 3 using ehci-pci
[620635.599518] usb 1-1: New USB device found, idVendor=2c7c, idProduct=6004
[620635.599519] usb 1-1: New USB device strings: Mfr=2, Product=3, SerialNumber=4
[620635.599767] usb 1-1: Product: EG060V_EAAA
[620635.599768] usb 1-1: Manufacturer: Quectel
[620635.599769] usb 1-1: SerialNumber: 123456789ABCD
[620635.648823] option 1-1:1.1: GSM modem (1-port) converter detected
[620635.650894] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB0
[620635.662098] option 1-1:1.2: GSM modem (1-port) converter detected
[620635.664149] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB1
[620635.665053] option 1-1:1.3: GSM modem (1-port) converter detected
[620635.667672] usb 1-1: GSM modem (1-port) converter now attached to ttyUSB2
[620635.676833] cdc_acm 1-1:1.4: ttyACM0: USB ACM device
[620635.970973] usbcore: registered new interface driver cdc_ether
[620635.995523] rndis_host: probe of 1-1:1.0 failed with error -16
[620635.995563] usbcore: registered new interface driver rndis_host
```

Figure 14: Loaded Module Ports

Step 3: Execute `./QLog -s log` in Linux terminal window to open QLog to capture log. When log capture completes, click “Ctrl” + “C” to end the QLog process.



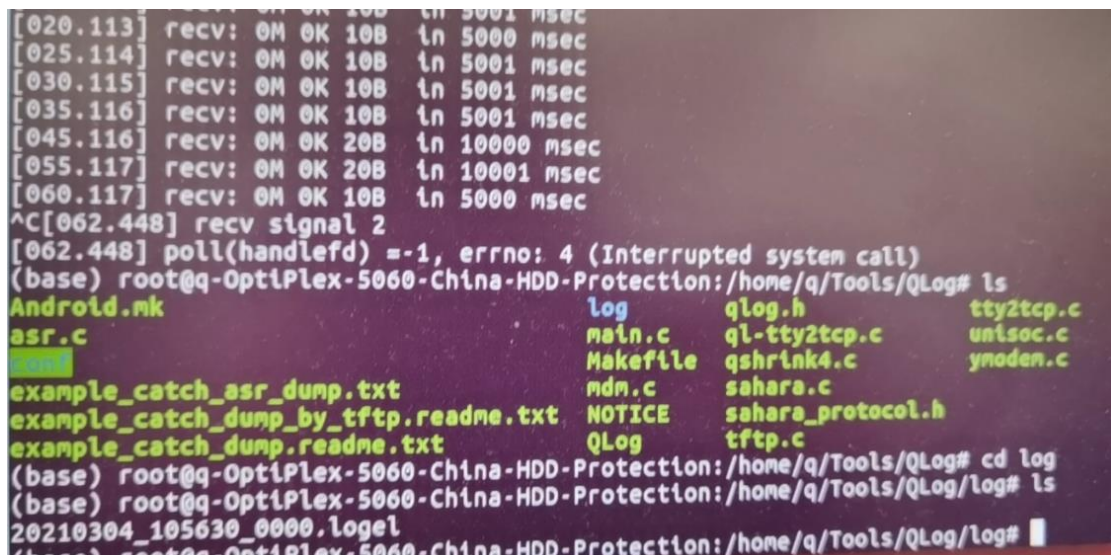
```

root@q-OptiPlex-5060-China-HDD-Protection: /home/q/Tools/QLog
q@q-OptiPlex-5060-China-HDD-Protection:~/Tools/QLog$ sudo su
[sudo] q 的密码:
(base) root@q-OptiPlex-5060-China-HDD-Protection:~/Tools/QLog# ./QLog -s log
[000.000] QLog Version: Quectel_QLog_Linux&Android_V1.4.10
[000.001] will save log into dir: log
[000.001] will use filter file: default filter
[000.102] Find [0] idVendor=2c7c, idProduct=0900, bNumInterfaces=8, ttyDM=/dev/ttyUSB0, busnum=002, dev=005, usbdevice_pah=/sys/bus/usb/devices/2-5
[000.102] ttyLOG=/dev/ttyUSB1
[000.102] open /dev/ttyUSB0 ttyfd = 3
[000.103] open /dev/ttyUSB1 ttyfd = 4
[000.110] Press CTRL+C to stop catch log.
[000.110] catch log via tty port
[000.110] unisoc_send_cmd cmd='0'
[000.110] qlog_logfile_create log/20210304_105630_0000.logel logfd=5
[005.112] recv: 0M 199K 617B in 5002 msec
[010.112] recv: 0M 0K 10B in 5000 msec
[015.113] recv: 0M 0K 10B in 5001 msec
[020.113] recv: 0M 0K 10B in 5000 msec
[025.114] recv: 0M 0K 10B in 5001 msec

```

Figure 15: Capture CP Log on Linux Device

Step 4: View the corresponding log file. You can save and send the log file to Quectel Technical Supports for further analysis. Take viewing the log file `20210304_05630_0000.logel` as an example:



```

[020.113] recv: 0M 0K 10B in 5001 msec
[025.114] recv: 0M 0K 10B in 5000 msec
[030.115] recv: 0M 0K 10B in 5001 msec
[035.116] recv: 0M 0K 10B in 5001 msec
[045.116] recv: 0M 0K 20B in 10000 msec
[055.117] recv: 0M 0K 20B in 10001 msec
[060.117] recv: 0M 0K 10B in 5000 msec
^C[062.448] recv signal 2
[062.448] poll(handlefd) = -1, errno: 4 (Interrupted system call)
(base) root@q-OptiPlex-5060-China-HDD-Protection:~/Tools/QLog# ls
Android.mk      log             qlog.h          tty2tcp.c
asr.c           main.c          ql-tty2tcp.c    unisoc.c
example_catch_asr_dump.txt  Makefile        qshrink4.c      ynodes.c
example_catch_dump_by_tftp.readme.txt  NOTICE         sahara.c
example_catch_dump.readme.txt           QLog            sahara_protocol.h
(base) root@q-OptiPlex-5060-China-HDD-Protection:~/Tools/QLog# cd log
(base) root@q-OptiPlex-5060-China-HDD-Protection:~/Tools/QLog/log# ls
20210304_105630_0000.logel
(base) root@q-OptiPlex-5060-China-HDD-Protection:~/Tools/QLog/log#

```

Figure 16: View CP Log on Linux Device

3.1.3. Capture CP Log Through QLog (Log Saved in Windows)

Step 1: Connect user's Linux CPE development board to Windows PC through LAN port, and configure a static address for the corresponding network port on the PC. For example, if the bridge address of the CPE development board is 192.168.1.x, then an address of the same network segment should be configured for the corresponding network port of the PC to ensure that the PC can ping the bridge address of the CPE development board.

Step 2: Run the QLog on the Linux CPE development board and export the log through the TCP port by executing `-s port` in the terminal window of Linux CPE development board. For example, `./Qlog-s 9000` means creating a TCP service at port 9000. Open CATStudio on Windows device and select "Socket" on "Communication Settings" interface. Select "TCP Client", enter the bridge address of the CPE development board and the specified port for running the log, then click "OK" to view the log capture on the "LogViewer" interface.

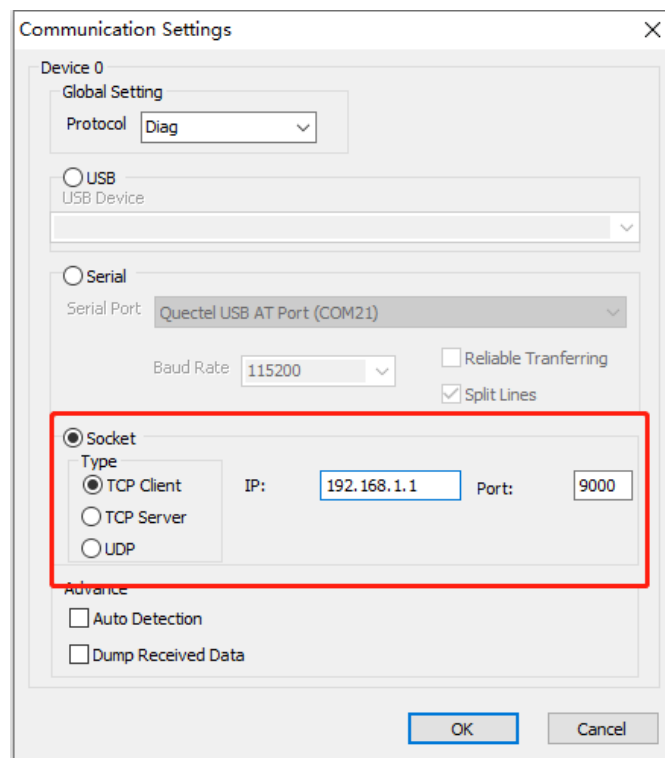


Figure 17: Socket Port Selection Interface

You can save and send the log to Quectel Technical Supports for further analysis. For the details of the step for saving the log, you can refer to **Step 6** of **Chapter 2.1.2**.

3.2. Capture Dump File

You can capture dump file through rdp_transfer in Linux. The module must be configured into dump mode firstly by executing **AT+QCFG="aprstlevel",0** and **AT+QCFG="modemrstlevel",0**. You have to reboot the module after **OK** is returned for both commands such that the commands can take effect. See **Chapter 4.2** for more details of the commands mentioned above.

In Linux system, when module dumps, the dump file needs to be exported through the LAN port of the Windows PC.

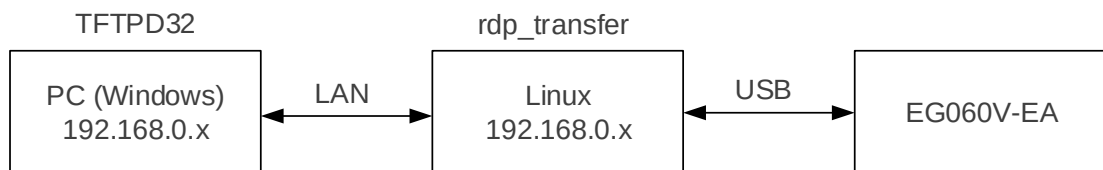


Figure 18: Connection Method of Capturing Dump File in Linux System

The steps for capturing dump file in Linux are as follows:

Step 1: Open the network management configuration interface on Windows PC and set up a private IP address for the LAN port network to connect the Linux device, for example, 192.168.0.10.

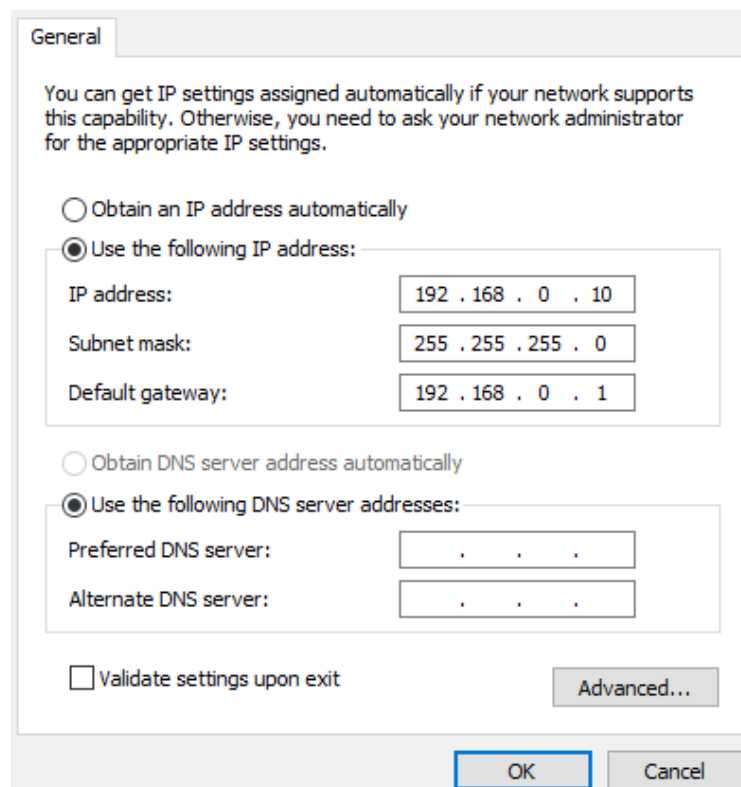


Figure 19: NIC Setting of Windows PC

Step 2: The Linux device (user's Linux device or Ubuntu) uses a configuration file to set a static address (which can only be modified when the root permission is obtained by entering "root" as user name and "admin" as password to log in to the Linux system of the module).

1) Modify the configuration file *interfaces* in the directory */etc/network/*, an example of which is illustrated as follows:

```
# interfaces(5) file used by ifup(8) and ifdown(8)
auto lo
iface lo inet loopback
auto eth0
iface eth0 inet static
address 192.168.0.3
netmask 255.255.255.0
gateway 192.168.0.1
```

2) Modify the configuration file *base* in the directory */etc/resolvconf/resolv.conf.d/*, an example of which is illustrated as follows:

```
nameserver 10.0.0.1
nameserver 192.168.0.1
```

After rebooting the device, you can execute **ifconfig** to check whether the eth0 network port is configured successfully, then **ping** the Linux device and Windows device to ensure the connectivity between the two devices.

Step 3: Test the module function and reproduce the dump problem. Firstly, compile the rdp_transfer source code into a rdp_transfer binary executable program on the host. The module automatically reboots after it dumps. Execute **sudo ./rdp_transfer < windows PC IP >** with root permission through the rdp_transfer process of the host and transmit the module's dump file to the Windows PC through the LAN port, and then export the log locally through TFTP32 of Windows PC.


```

Exiting dtagserver...
q@q-Latitude-5490:~/rdp_transfer$ sudo ./rdp_transfer 192.168.100.10
[sudo] password for q:
=====
ASR RDP Transfer v1.0.0.0
=====
Found USB Devices:
  DevNum      Path                      iSerialNumber
-----
# Device 0: /dev/bus/usb/001/014  123456789ABCD
nSelectedDev: 0
copy serial number
copy serial number done
manual srv ip: 192.168.100.10
srv ip: 192.168.100.10
detect tftp server
recv OP_ERROR from pc, server ok
port: 63707
connect usbrdp device ok
claim interface return 0
main-495: notify RD-begin
Write 11 bytes to UE!
main-507: waiting for RDP_TYPE
main-518: RDP_TYPE done
>>>>>>>>>>Goto CP_ASSERT flow

main-911: get dump req, sending CPASSERT_LOG_2020_12_28_19_21_35.tar.gz file to srv
send file name:CPASSERT_LOG_2020_12_28_19_21_35.tar.gz len: 48
cpassert fname ACK block number: 0
Write 13 bytes to UE!
=====dump0 CPASSERT_LOG_2020_12_28_19_21_35.tar.gz done
total send bytes: 0xd3b6a

longing to catch another dump image, please waiting....
main-911: get dump req, sending CPASSERT_COMDDR_2020_12_28_19_21_35.bin file to srv
send file name:CPASSERT_COMDDR_2020_12_28_19_21_35.bin len: 48
cpassert fname ACK block number: 0
Write 13 bytes to UE!
ACK block number: 20000
total send bytes: 0x9c3e00
=====dump1 CPASSERT_COMDDR_2020_12_28_19_21_35.bin done
total send bytes: 0x11d6ffc
  
```

Figure 20: Printed Information After Successful Capture of Log by rdp_transfer

4 Dump Mode Related AT Commands

4.1. AT Command Introduction

4.1.1. Definitions

- **<CR>** Carriage return character.
- **<LF>** Line feed character.
- **<...>** Parameter name. Angle brackets do not appear on the command line.
- **[...]** Optional parameter of a command or an optional part of TA information response. Square brackets do not appear on the command line. When an optional parameter is not given in a command, the new value equals to its previous value or the default settings, unless otherwise specified.
- **Underline** Default setting of a parameter.

4.1.2. AT Command Syntax

All command lines must start with **AT** or **at** and end with **<CR>**. Information responses and result codes always start and end with a carriage return character and a line feed character: **<CR><LF><response><CR><LF>**. In tables presenting commands and responses throughout this document, only the commands and responses are presented, and **<CR>** and **<LF>** are deliberately omitted.

Table 1: Types of AT Commands

Command Type	Syntax	Description
Test Command	AT+<cmd>=?	Test the existence of corresponding Write Command and return information about the type, value, or range of its parameter.
Read Command	AT+<cmd>?	Check the current parameter value of a corresponding Write Command.
Write Command	AT+<cmd>=<p1>[,<p2>[,<p3>[...]]]	Set user-definable parameter value.
Execution Command	AT+<cmd>	Return a specific information parameter or perform a specific action.

4.2. AT Command Description

4.2.1. AT+QCFG="aprstlevel" Set AP Dump Level

This command queries or sets AP dump level.

AT+QCFG="aprstlevel" Set AP Dump Level	
Write Command AT+QCFG="aprstlevel"[,<level>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "aprstlevel",<level> OK If the optional parameter is specified, set AP dump level: OK If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module is rebooted. The configuration will be saved automatically.

Parameter

<level>	Integer type. AP dump level.
	0 Open AP dump. Module restarts and enters dump mode when AP dump happens
	1 Close AP dump. Module only restarts when AP dump happens

4.2.2. AT+QCFG="modemrstlevel" Set CP Dump Level

This command queries sets CP (corresponding to modem) dump level.

AT+QCFG="modemrstlevel" Set CP Dump Level	
Write Command AT+QCFG="modemrstlevel"[,<level>]	Response If the optional parameter is omitted, query the current setting: +QCFG: "modemrstlevel",<level> OK If the optional parameter is specified, set modem dump level: OK

	If there is any error: ERROR
Maximum Response Time	300 ms
Characteristics	The command takes effect after the module is rebooted. The configuration will be saved automatically.

Parameter

<level>	Integer type. Modem dump level.
0	Open modem dump. Module restarts and enters dump mode when CP dump happens
1	Close modem dump. Only CP restarts when CP dump happens

NOTE

If it is required that the module can enter dump mode after either AP dump happens or CP dump happens, then the module has to be rebooted after **AT+QCFG="aprstlevel",0** and **AT+QCFG="modemrstlevel",0** are both executed successfully.

5 Network State Checking Related AT Commands

The AT commands for checking the network state of the module are listed as follows:

1. **AT+CFUN?** Query the function mode of the module
2. **AT+CPIN?** Query the status of SIM card
3. **AT+CEREG?** Query EPS network registration status
4. **AT+COPS?** Query network status
5. **AT+QENG="servingcell"** Query serving cell information
6. **AT+CGDCONT?** Query PDP configuration
7. **AT+CGPADDR** Query PDP address
8. **AT+CGACT?** Query PDP activation status

NOTE

For more details about the AT commands mentioned above, please refer to **document [1]**.

6 Appendix References

Table 2: Related Documents

Document Name
[1] Quectel_EG060V-EA_AT_Commands_Manual

Table 3: Terms and Abbreviations

Abbreviation	Description
AP	Application Processor
API	Application Programming Interface
CP	Central Processor
CPE	Customer-Premise Equipment
ECM	Ethernet Control Model
EPS	Evolved Packet System
IP	Internet Protocol
LAN	Local Area Network
PC	Personal Computer
PDP	Packet Data Protocol
PIN	Personal Identification Number
RNDIS	Remote Network Driver Interface Specification
SIM	Subscriber Identity Module
TA	Terminal Adapter
TCP	Transmission Control Protocol

TFTP	Trivial File Transfer Protocol
UART	Universal Asynchronous Receiver/Transmitter
USB	Universal Serial Bus